ELECTRICITY -01

Class 10 - Science

1. A metallic conductor has loosely bound electrons called free electrons. The metallic conductor is: **[1]**

a) Positively charged
b) Negatively charged
c) Neutral
d) Either positively charged or negatively charged

2. The values of mA and μA are **[1]**

a) 10^{-3} A and 10^{-9} A respectively
b) 10^{-6} A and 10^{-3} A respectively
c) 10^{-6} A and 10^{-9} A respectively
d) 10^{-3} A and 10^{-6} A respectively

3. A current of 1 A is drawn by a filament of an electric bulb. Number of electrons passing through a cross section of the filament in 16 seconds would be roughly **[1]**

a) 10^{23}
b) 10^{16}
c) 10^{18}
d) 10^{20}

4. Which unit could be used to measure current? **[1]**

a) Watt
b) Coulomb
c) Ampere
d) Volt

5. An ammeter has 20 divisions between mark 0 and mark 2 on its scale. The least count of the ammeter is **[1]**

a) 0.01 A
b) 0.02 A
c) 0.2 A
d) 0.1 A

6. Which of the following represents voltage? **[1]**

a) $\frac{\text{Work done}}{\text{Current} \times \text{Time}}$
b) $\frac{\text{Work done} \times \text{Time}}{\text{Current}}$
c) Work done $\times$ Charge $\times$ Time
d) Work done $\times$ Charge

7. The unit for measuring potential difference is: **[1]**

a) kWh
b) Volt
c) Ohm
d) Watt

8. Match the column I with column II and select the correct option from the codes given. **[1]**

Column I	**Column II**
(a) Electric current	(i) volt

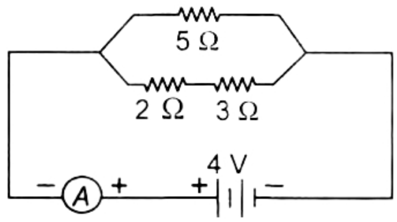
(b) emf	(ii) ohm
(c) Resistance	(iii) ohm-metre
(d) Resistivity	(iv) ampere

a) (a) - (iii), (b) - (i), (c) - (ii), (d) - (iv)
b) (a) - (iv), (b) - (ii), (c) - (i), (d) - (iii)
c) (a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)
d) (a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)

9. The work done in moving a unit charge across two points in an electric circuit is a measure of: **[1]**

a) Potential difference
b) Power
c) Resistance
d) Current

10. What is the potential difference across 2Ω in the given circuit? **[1]**



a) 1 V
b) 1.6 V
c) 5 V
d) 2.5 V

11. The device used for measuring potential difference is known as: **[1]**

a) Voltmeter
b) Galvanometer
c) Ammeter
d) Potentiometer

12. In a voltmeter, there are 20 divisions between the 0 mark and 0.5 V mark. The least count of the voltmeter is **[1]**

a) 0.050 V
b) 0.250 V
c) 0.025 V
d) 0.020 V

13. Work done to move 1 coulomb charge from one point to another point in a charged conductor having a potential 10 volt is**:** **[1]**

a) 1 Joule
b) 100 Joule
c) 10 Joule
d) Zero

14. An electric fuse works on the: **[1]**

a) chemical effect of current
b) lighting effect of current
c) heating effect of current
d) magnetic effect of current

15. If 20C of charge pass a point in a circuit in 1s, what current is flowing? **[1]**

a) None of these
b) 30A
c) 20A
d) 20C

16. A voltmeter is always connected in parallel because it has **[1]**

a) Very low resistance
b) High power rating
c) Very high resistance
d) No resistance

17. If the amount of electric charge passing through a conductor in 10 minutes is 300 C, the current flowing is: **[1]**

a) 5 A
b) 30 A
c) 0.5 A
d) 0.3 A

18. A current of 4 A flows around a circuit for 10s. How much charge flows past a point in the circuit in this time? **[1]**

a) 40 C
b) 4 A
c) None of these
d) 4 C

19. How many electrons are flowing per second past a point in a circuit in which there is a current of 5A? **[1]**

a) 31.25×10^{17}C
b) 31.25×10^{19}C
c) 31.25×10^{18}C
d) None of these

20. What is the current in a circuit if the charge passing each point is 20 C in 40 s? **[1]**

a) 0.5 C
b) 0.5 A
c) None of these
d) 0.2 C

21. The slope of V-I graph gives **[1]**

a) Potential difference
b) Current
c) Conductivity
d) Resistance

22. Which of the following obeys Ohm's law? **[1]**

a) Nichrom
b) Filament of a bulb
c) LED
d) Transistor

23. An electrical appliance has a resistance of 25 Ω. When this electrical appliance is connected to a 230 Ω supply line, the current passing through it will be: **[1]**

a) 92 A
b) 9.2 A
c) 2.9 A
d) 0.92 A

24. The nature of the graph between potential difference and the electric current flowing through a conductor is **[1]**

a) hyperbolic
b) parabolic
c) circle
d) straight line

25. A potential difference of 10V is needed to make a current of 0.02A flow through a wire. What potential difference is needed to make a current of 250mA flow through the same wire? **[1]**

a) 135 volt
b) None of these
c) 125 volt
d) 25 volt

26. While performing the experiment to study the dependence of current on potential difference, if the circuit used to measure the current and voltage is kept on a position for a longer time, then: **[1]**

a) All of these
b) Ammeter reading will change
c) The resistor will get heated up changing the value of "R"
d) Voltmeter reading will change

27. If the charge on an electron is 1.6×10^{-19} coulombs, how many electrons should pass through a conductor in 1 **[1]**

second to constitute 1 ampere current?

a) 6.25×10^{-18}

b) 6.35×10^{-18}

c) None of these

d) 6.25×10^{-19}

28. A resistance of 20 ohms has a current of 2 amperes flowing in it. What potential difference is there between its ends? **[1]**

a) None of these

b) 20 V

c) 35 V

d) 40 V

29. How much energy is consumed when a current of 5 ampere flows through the filament of a heater having a resistance of 100 ohms for two hours? Express it in joules. **[1]**

a) 18×10^5 J

b) 18×10^6 J

c) 18×10^7 J

d) None of these

30. The current passing through a conductor is: **[1]**

a) Inversely proportional to the power

b) Inversely proportional to the potential difference

c) Directly proportional to the potential difference

d) Directly proportional to the resistance

31. Ohm’s law gives a relation between: **[1]**

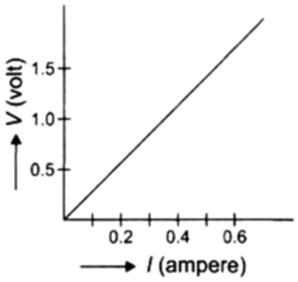
a) Resistance and potential difference

b) Potential difference and electric charge

c) Current and potential difference

d) Current and resistance

32. Following graph was plotted between V and I values, across a metal wire. Which of the following statement(s) is/are correct regarding this? **[1]**



a) This graph illustrates the non-ohmic law.

b) While plotting this graph, the temperature remains constant.

c) Value of ratio $\frac{V}{1}$ when the potential difference in 0.8 V is not equal to the value of ratio $\frac{V}{1}$ when the potential difference is 1.2 V.

d) All of these

33. A car headlight bulb working on a 12V car battery draws a current of 0.5 A. The resistance of the light bulb is: **[1]**

a) 0.5 Ω

b) 6 Ω

c) 12 Ω

d) 24 Ω

34. There is a dual of 8 ohm resistance on the aerial. Determine the aerial's new resistance. **[1]**

a) 10 Ω | b) 2 Ω
c) 7 Ω | d) 4 Ω

35. The resistivity of certain material is 0.6 Ωm. The material is most likely to be: **[1]**

a) A conductor
b) A semiconductor
c) A superconductor
d) An insulator

36. Which of the following instruments does not have plus (+) or minus (-) sign marked on it while representing in a circuit diagram? **[1]**

a) Galvanometer
b) Ammeter
c) Cell
d) Rheostat

37. The potential difference across a 3Ω resistor is 6V. The current flowing in the resistor will be: **[1]**

a) 6 A
b) 2 A
c) 1 A
d) $\frac{1}{2}$ A

38. When a 4 Ω resistor is connected across the terminals of a 12V battery, the number of coulombs passing through the resistor per second is: **[1]**

a) 0.3
b) 3
c) 4
d) 12

39. The unit of electrical resistance is: **[1]**

a) Ampere
b) Coulomb
c) Ohm
d) Volt

40. Let us consider the current flowing through a metallic wire if the temperature of the entire system increases. What will happen from the following options? **[1]**

a) Resistance (R) decreases
b) Potential difference (V) decreases
c) Potential difference (V) increases
d) V and R remains the same

41. The value for the SI unit of electrical potential difference can be calculated using which of the following formula? **[1]**

a) $\frac{\text{1 Joule}}{\text{1 Coulomb}}$
b) 1 Joule $\times$ 1 Coulomb
c) $\frac{\text{1 Volt}}{\text{1 Ampere}}$
d) 1 volt $\times$ 1 Ampere

42. An electric bulb is rated at 220V, 100W. What is its resistance? **[1]**

a) 454 ohm
b) None of these
c) 254 ohm
d) 484 ohm

43. How many electrons constitute one coulomb of charge? **[1]**

a) 6.18×10^{18}
b) 2.25×10^{18}
c) 6.25×10^{18}
d) 6.25×10^{19}

44. A current of 200mA flows through a 4 kΩ resistor. What is the potential difference across the resistor? **[1]**

a) 4000 volt
b) 900 volt

c) 800 volt | d) None of these

45. If the current through a flood lamp is 5A, what charge passes in 10 seconds? **[1]**

a) 50 C
b) 2 C
c) 5 C
d) 0.5 C

46. One coulomb charge is equivalent to the charge obtained in: **[1]**

a) 6.2×10^{19} electrons
b) 6.25×10^{18} electrons
c) 2.6×10^{19} electrons
d) 2.65×10^{18} electrons

47. A current of 5 amperes flows through a wire whose ends are at a potential difference of 3 volts. The resistance of the wire: **[1]**

a) 0.7 Ohms
b) 0.1 Ohms
c) 0.6 Ohms
d) None of these

48. Current flows through a conductor connected across a voltage source. Now the resistance of the conductor is reduced to one fourth to its initial value and connected across the same voltage source. The heating effect in the conductor will become **[1]**

a) One fourth
b) Four times
c) Double
d) Half

49. An electric heater is rated at 1500 W. How much heat is produced per hour? **[1]**

a) 5.4×10^{6} J
b) 5400 J
c) 5.4×10^{5} J
d) 54000 J

50. Two bulbs of 100 W and 40 W are connected in series. The current through the 100 W bulb is 1 A. The current through the 40 W bulb will be: **[1]**

a) 1 A
b) 0.4 A
c) 0.8 A
d) 0.6 A

51. An electric room heater draws a current of 2.4A from the 120V supply line. What current will this room heater draw when connected to 240V supply line? **[1]**

a) 5.8 amp
b) 4.9 amp
c) None of these
d) 4.8 amp

52. The wire connected necessarily in series is **[1]**

a) Fuse wire
b) Connecting wire
c) Heating wire
d) Source wire

53. An electric kettle consumes 1 kW of electric power when operated at 220 V. A fuse wire of what rating must be used for it? **[1]**

a) 4 A
b) 1 A
c) 5 A
d) 2 A

54. The current passing through an electric kettle has been doubled. The heat produced will become: **[1]**

a) four times | b) one-fourth
c) double | d) half

55. An electric fan runs from the 230V mains. The current flowing through it is 0.4A. At what rate is electrical energy transferred by the fan? **[1]**

a) 91 J/s | b) None of these
c) 92 J/s | d) 95 J/s

56. The elements of electric heating devices are usually made of: **[1]**

a) nichrome | b) bronze
c) argon | d) tungsten

57. In an electrical circuit two resistors of 2Ω and 4Ω respectively are connected in series to a 6 V battery. The heat dissipated by the 4Ω resistor in 5 s will be **[1]**

a) 30 J | b) 20 J
c) 10 J | d) 5 J

58. An electric kettle for use on a 230 V supply is rated at 3000 W. For safe working, the cable connected to it should be able to carry at least: **[1]**

a) 5 A | b) 15 A
c) 2 A | d) 10 A

59. If the current I through a resistor is increased by 100% (at constant temperature), the increase in power dissipated will be **[1]**

a) 400% | b) 300%
c) 100% | d) 200%

60. A resistance of 25 Ω is connected to a 12 V battery. The heat energy in joules generated per minute: **[1]**

a) 300 J | b) 347.6 J
c) 345.6 J | d) None of these

61. A heating coil has a resistance of 200Ω. At what rate will the heat be produced in it when a current of 2.5 A flows through it? **[1]**

a) 1250 J/s | b) 1200 J/s
c) None of these | d) 600 J/s

62. If the current flowing through a fixed resistor is halved, the heat produced in it will become: **[1]**

a) one-fourth | b) double
c) four times | d) one-half

63. An electric heater is rated at 2kW. Electrical energy costs rupees 4 per kWh. What is the cost of using the heater for 3 hours? **[1]**

a) Rupees 12 | b) Rupees 36
c) Rupees 24 | d) Rupees 48

64. The commercial unit of energy is: **[1]**

a) Kilo-joule
b) Kilowatt-hour
c) Watt-hour
d) Watt

65. 100 joules of heat is produced per second in a 4 ohm resistor. What is the potential difference across the resistor? **[1]**

a) 50 V
b) None of these
c) 20 V
d) 10 V

66. An electric fuse is connected with: **[1]**

a) Live wire
b) Earthing
c) Neutral wire
d) Parallel to the line wire

67. In a house, two 60W electric bulbs are lighted for 4 hours and three 100W bulbs for 5 hours every day. The electric energy is consumed in 30 days: **[1]**

a) 59.4 kWh
b) None of these
c) 100 kWh
d) 45 kWh

68. The heat produced in a wire of resistance **x** when a current **y** flows through it in time **z** is given by: **[1]**

a) $y \times z \times x$
b) $x^2 \times y \times z$
c) $x \times z \times y^2$
d) $y \times z^2 \times x$

69. The resistance of a resistor is reduced to half of its initial value. If other parameters of the electrical circuit remain unaltered, the amount of heat produced in the resistor will become: **[1]**

a) half
b) two times
c) four times
d) one fourth

70. An air conditioner is rated 260 V, 2.0 kW. The air conditioner is switched on for 10 hours each day. What is electrical energy consumed in 30 days? **[1]**

a) 600 kW h
b) 420 kW h
c) 20 kW h
d) 2000 kW h

71. The SI unit of energy is: **[1]**

a) Ohm-meter
b) Joule
c) Watt
d) Coulomb

72. An electric heater of resistance 8 Ω takes a current of 15 A from the mains supply line. The rate at which heat is developed in the heater: **[1]**

a) 1500 J/s
b) None of these
c) 1800 J/s
d) 1600 J/s

73. Which of the following characteristics is not suitable for a fuse wire? **[1]**

a) Higher resistance than the rest of the wiring
b) Thick and short
c) Thin and short
d) Low melting point